

Marcos María Martínez Varas

Junior Data Engineer · Python · SQL · Generative AI

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> **Availability:** immediate, full-time.

PROFILE

Higher Technician in Multiplatform Application Development (DAM) with a 4-month internship at Minsait (Indra Group), currently studying a Master's in Big Data, Applied AI and Generative AI at U-tad. I build data and generative AI projects with Python, SQL and automated testing, on a full-stack foundation in Java/Spring Boot and Angular. English C2 (LanguageCert). Looking for an internship or first role as a junior data professional.

EXPERIENCE

Minsait (Indra Group) · Software Development Intern — Mar – Jun 2026 (4 months)

- Front-end development with Angular, Tailwind CSS and JavaScript as part of a professional development team.
- Back-end tasks with Java and Spring Boot.
- Worked within the team's real workflow: version control with Git, reviews and task-based delivery.

EDUCATION

Master's in Big Data, Applied AI and Generative AI · U-tad — 2026 – 2027 (in progress)

Higher Technician in Multiplatform Application Development (DAM) · IES Gonzalo Chacón — 2024 – 2026

PROJECTS

Code and documentation at github.com/mmartinezvaras

RAG assistant for PDFs · Python, LangChain, ChromaDB, Ollama, Streamlit
Answers questions about PDF documents with a local LLM, citing the document and page behind every answer. LLM-as-a-judge evaluation with pytest on GitHub Actions for every change; one-click demo with Codespaces and Docker.

Banking lakehouse · Python, PySpark, Delta Lake · **In progress**
Reproducible simulated bank: 1,000 customers and 50,062 transactions with 62 frauds, tested on GitHub Actions. In progress: Bronze/Silver/Gold pipeline with PySpark and Delta Lake, and fraud detection with window functions.

LaLiga match prediction · Python, pandas, scikit-learn
Random Forest with time-based validation and leakage-free recent-form features: 51.5% precision when predicting wins, against a 38% base rate.

Resale management dashboard · Java 21, Spring Boot, MySQL, Angular
REST API with 64 endpoints, pessimistic locking to keep stock consistent, an auditable inventory ledger and 26 tests with JUnit 5 and Mockito.

NoSQL banking database · MongoDB Atlas, PyMongo
Customers modelled as documents with embedded data, single and compound indexes, and aggregation pipelines that compute KPIs such as the average balance per account type.

TECHNICAL SKILLS

Data: Python, SQL, pandas, MySQL, MongoDB · Spark/PySpark (basic) · Hadoop HDFS (basic)
Generative AI and ML: LLMs, RAG, LangChain, ChromaDB, Ollama, prompting, scikit-learn
Development: Java, Spring Boot, JPA, Angular, TypeScript, JavaScript, Tailwind CSS, HTML/CSS
Quality and DevOps: Git, GitHub Actions, Docker, pytest, JUnit 5
Mainframe: COBOL (learning)

LANGUAGES AND OTHER

Languages: Spanish (native) · English C2 (LanguageCert)
Other: Driving licence · Football team coach (leadership and organisation)